# Dynamic Discrete Choice and Inverse Reinforcement Learning: Inferring preferences and beliefs from human behavior*

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## 1 Introduction

We survey two related literatures: *dynamic discrete choice* (DDC) in structural econometrics (SE) and "inverse reinforcement learning" (IRL) in machine learning (ML). Both are focused on inferring preferences of individuals from observations of their choices over time assuming they are *rational* i.e. their choices maximize an expected reward ("utility") function based on beliefs of how their actions affect current and uncertain future states, outcomes, and payoffs, allowing for unobserved states ("noise") that result in decision rules that appear to be probabilistic from the standpoint of an outside observer who only partially observes agents' states, Rust (1987). The literatures differ in the methods used to solve the decision problem and infer agents' preferences, and have different rationales for inferring/learning them.

SE focuses on testing and evaluating economic theories and making *counterfactual predictions* of the effect of changes in the environment and economic policies on economic behavior, outcomes, and welfare.[1] Structural models differ from reduced-form models that summarize behavior without inferring preferences. Reduced-form models require historical variation in policies that allow them to be used as covariates to make counterfactual predictions. Structural models can simulate how agents would react to novel policies that have no close historical antecedents. For example, Denmark has long had one of the world's highest tax rates on new cars, but with little historical variation that could be used by reduced form approaches to predict how reductions in the tax would affect the Danish economy. Gillingham et al. (2022) structurally estimated an equilibrium model of ownership and trading of cars to recover consumer preferences under the high tax *status quo* regime, and used it to compute counterfactual equilibria under a variety of alterative tax rates. They showed that by reducing the tax on new cars and increasing the gas tax, Denmark can: 1) improve the welfare of most of its citizens, 2) increase total tax revenue from cars, and 3) reduce total $CO_2$ pollution. Credible structural models are a faster, cheaper way of predicting the effects of policy changes by "crash testing" computer simulations of new policies before actually implementing them.

IRL, introduced by Ng and Russell (2000), builds on *reinforcement learning* (RL) (Sutton and Barto, 1998). Its goal is to develop intelligent agents that can perform complex robotic and intellectual tasks without the need for explicit programming or an explicit reward function to direct behavior. The success of ML is due to *supervised learning* where algorithms are trained to predict outcomes using labeled data. However supervised learning is data-intensive and does not naturally handle sequential decision making where actions have long-term consequences. RL overcomes this by training agents to maximize a reward through trial-and-error, via repeated dynamic interactions with the environment. When the reward function is known, RL can approximate optimal behavior without labeled examples. Chess is an example with a known reward (1 for a win, -1 for a loss and 0 for a draw). RL has been able to learn superhuman performance via *self-play,* i.e. repeatedly playing the algorithm against itself resulting in performance that exceeds the best human and

[1] In economics "policy" refers to government interventions (tax rates, regulations, etc.), whereas in RL/MDP it refers to a decision rule mapping states to actions.

computerized chess players, see e.g. Silver et al. (2018).

Christiano et al. (2017) note that it is unclear what reward function to use to program tasks in AI and robotic application using RL, for example in trying to "train a robot to clean a table or scramble an egg." IRL solves this by learning a reward function from limited data on human or expert demonstrations. Using the inferred reward function, these algorithms can be trained more rapidly and cheaply via RL using computer simulations that result in intelligent behavior consistent with human preferences in situations outside of those in a limited demonstration data set. For example, Google Maps trip data (Barnes et al., 2024) revealed travelers' route preferences reflecting trade-offs between traffic, distance, hills, safety, and scenery. They massively scaled IRL to estimate a route preference function (360M parameters, 110M trips) enabling RL to generate recommendations achieving 16-24% improvement in global route quality, and to the best of our knowledge, represents the largest published study or IRL algorithms in a real-world setting to date.

Section 2 reviews Dynamic Programming (DP) and Markovian Decision Processes (MDP), the common framework used in the DDC and IRL literatures, and reinforcement learning (RL). Section 3 reviews the DDC literature and section 4 the IRL literature. Section 5 concludes, comparing/contrasting the literatures and the key challenges and limitations they both face.

## 2 Background on DP and MDPs

DDC and IRL apply the recursive mathematical characterization of optimal sequential decision making under uncertainty known as *dynamic programming* (DP), Bellman (1957).[2] We focus on stationary, infinite horizon Markovian Decision Problems (MDP), a subclass of DPs introduced by Bellman and Howard (1960) that are the dominant paradigm in SE and RL. MDPs are defined by 1) a state space $S$, 2) an action space $A$, and 3) the objects $\{r, p, \beta\}$, where $r(s, a)$ is the *reward or utility function* providing the payoff to a decision maker (DM), $p(s'|s, a)$ is a *transition density* for next period's state when the state is $s$ and action is $a$ and $\beta \in (0, 1)$ is a discount factor. Assume that $S$ and $A$ are finite sets with $|S|$ and $|A|$ elements, respectively. A policy or *decision rule* $\pi$ is a conditional probability of choosing a feasible action $a \in A(s)$ for all $s \in S$. It is called a *mixed strategy* in game theory to distinguish it from a *pure strategy,* i.e. a deterministic function $a = \delta(s)$ that chooses a single optimal action with probability 1. Since mixed strategies include pure strategies as a special case, we can express the DM's objective as the choice of a policy $\pi$ that maximizes expected discounted rewards given by

$$V(s) = \max_{\pi} V_{\pi}(s), \quad \text{where } V_{\pi}(s) = E_{\pi}\left\{\sum_{t=0}^{\infty} \beta^t r(s_t, a_t)|s_0 = s\right\}, \tag{1}$$

where $E_{\pi}$ denotes the expectation operator over the Markov process $\{s_t, a_t\}$ implied by the transition probability $p$ and policy $\pi$. The optimal value function can be defined recursively via *Bellman's*

[2]See Bertsekas (2017) for an excellent introduction to the DP literature.

*equation*

$$V(s) = \max_{a \in A(s)} \left[ r(s,a) + \beta \sum_{s' \in S} V(s') p(s'|s,a) \right] \equiv \Gamma(V)(s). \tag{2}$$

$V$ is the fixed point $V = \Gamma(V)$ to an operator $\Gamma(V)$ known as the "Bellman operator" which can be shown to be a *contraction mapping* (i.e. for any vectors $V, W \in R^{|S|}$ we have $\|\Gamma(V) - \Gamma(W)\| \leq \beta \|V - W\|$ where $\beta$ is less than 1 and $\| \cdot \|$ is the Euclidean norm). This implies that a unique solution $V$ to Bellman's equation exists, and the optimal policy $\pi$ takes the form of a pure strategy, i.e. a deterministic function $a = \delta^*(s)$ that can be recovered from $V$ by

$$\delta^*(s) = \underset{a \in A(s)}{argmax} \left[ r(s,a) + \beta \sum_{s' \in S} V(s') p(s'|s,a) \right], \tag{3}$$

since generically a unique action $a^* = \delta^*(s)$ attains the max on the right hand side of (3). It follows that any policy $\pi$ that fails to choose the optimal action with probability 1 is suboptimal. We now summarize the key methods for solving for $V$ and $\delta^*$.

### 2.1 Successive Approximations

Since a contraction mapping $\Gamma$ has a unique fixed point $V = \Gamma(V)$, the contraction property guarantees that the sequence $\{V_k\}$ given by $V_k = \Gamma(Vi_{k-1})$, for $k = 1, 2, \ldots$ starting from any initial starting guess $Vi_0$ converges to the fixed point $V = \Gamma(V)$ at a geometric rate in $\beta$, which is very slow when $\beta$ is close to 1.

### 2.2 Policy Iteration

The Howard (1960) *policy iteration* (PI) algorithm is a faster method for solving MDPs that alternates between *policy valuation* and *policy improvement* steps. At iteration $k$ the PI algorithm has a candidate policy $\delta_k$. The policy valuation step calculates the value of $\delta^k$, $V_{\delta^{(k)}}$, by solving the linear system

$$V_{\delta_k} = r_{\delta_k} + \beta E_{\delta_k} V_{\delta_k}, \tag{4}$$

where $r_{\delta_k}$ is the vector whose $s^{th}$ element is $r(s, \delta_k(s))$ and $E_{\delta_k}$ is the $|S| \times |S|$ matrix whose element in row $s$ and column $s'$ is given by $E_{\delta_k}(s, s') = p(s'|s, \delta_k(s))$. For any feasible deterministic policy $\delta$ the matrix $E_\delta$ is a Markov transition probability matrix with matrix norm $\|E_\delta\| = 1$, which implies that equation (4) has a unique solution given by $V_{\delta_k} = [I - \beta E_{\delta_k}]^{-1} r_{\delta_k}$. Given the value $V_{\delta_k}$ the policy improvement step calculates a new decision rule $\delta_{k+1}$ given by the right hand side of equation (3) except with $V_{\delta_k}$ substituted for $V$. If $\delta_{k+1} = \delta_k$ then policy iteration has converged and it is not hard to show that $V_{\delta_k} = V$, the unique solution to Bellman's equation (2).

### 2.3 Randomization, function approximation, and the Curse of Dimensionality

SE and IRL repeatedly solve MDPs to find a reward function whose implied optimal behavior best approximates observed behavior according to different metrics of goodness of fit. It is already challenging to solve a single MDP when $|S|$ or $|A|$ is large since each SA or PI step requires $O(|A||S|^2)$ arithmetic operations to compute the expectation and maximization operators in equation (2) for all $s \in S$. Though PI requires only a few iterations to converge and thus can be faster for smaller problems when $\beta$ is near 1, the downside is the $O(|S|^3)$ operations required to solve the linear system (4) in each policy valuation step. In the absence of sparsity or other "special structure" alternative methods are required when $|S|$ or $|A|$ is large.

Bellman called this the *curse of dimensionality* because it arises when discretization and interpolation are used to approximate solutions to MDPs with mulitple continuous state variables. Approximating a continuous MDP with $d$ state variables using a finite grid of $N$ points in each dimension results in an approximate finite MDP with $|S| = O(N^d)$ states which grow exponentially in $d$. The curse of dimensionality is unavoidable (Chow and Tsitsiklis, 1989): MDPs with $d_s$ continuous state and $d_c$ continuous choice variables have a worst-case complexity (minimum number of arithmetic operations to compute an $\varepsilon$-approximation to $V$) that is bounded below by a function that grows at rate $1/[(1-\beta)\varepsilon)]^{2d_s+d_c}$, a lower bound that holds for *any possible algorithm.*

It may be possible to break the curse of dimensionality for subclasses of MDPs with special structure, such as DDCs where the choice set $A$ is finite. When $|A|$ is small, most of the work to solve the MDP is spent on 1) numerical integration to compute conditional expectations $EV(s,a) = \int V(s')p(s'|s,a)$, and 2) interpolation/approximation needed to find $V(s')$ for $s' \in S$. While randomization techniques such as Monte Carlo integration have been proposed to mitigate the curse of dimensionality (Rust, 1997), their general applicability remains debated (Bray, 2022).

Another way to break the curse of dimensionality is to approximate $V$ by a function $V_\phi$ that depends on a small number of unknown parameters $\phi$ for which $V_\phi$ approximately solves the Bellman equation. For example $V_\phi$ could be the linear combination $V_\phi(s) = \sum_{k=1}^K \phi_k \rho_k(s)$ where $\{\rho_k\}$ are "basis functions" and $\phi = (\phi_1, \ldots, \phi_K)$ are coefficients, or $V_\phi$ could be a nonlinear function of $\phi$ such as a neural network. Neural networks are universal approximators, which might help break the curse of dimensionality of approximating $V$, at least for certain subclasses of sufficiently smooth functions as first demonstrated by Barron (1993). Let $V_{\hat{\phi}}$ denote a best approximation to $V$, where $\hat{\phi}$ minimizes the sum of squared "Bellman residuals" over a grid of points $(s_1, \ldots, s_J)$ in $S$

$$\hat{\phi} = \underset{\phi}{argmin} \sum_{j=1}^{J} [V_\phi(s_j) - \Gamma(V_\phi)(s_j)]^2. \tag{5}$$

Neural net approximation breaks the curse of dimensionality by allowing us to approximate $V$ with a small number of network weights/parameters $\phi$. Randomization also helps break the curse of dimensionality when we use Monte Carlo integration instead deterministic quadrature to compute

the conditional expectation entering the Bellman updates, $\Gamma(V_\phi)(s_j)$, and avoid numerical integration to calculate the mean squared Bellman error over the state space $S$ in (5). This transforms the solution of the Bellman equation for $V$ into a type of nonlinear regression problem.

Unfortunately, even if $\phi$ has relatively few elements and we can evaluate the Bellman errors in (5) rapidly, there may still be a curse of dimensionality associated with finding a global mimimizer $\hat{\phi}$.[3] Further, a $V_{\hat{\phi}}$ that approximately minimizes the sum of squared Bellman errors (5) need not be close to the true $V$ that solves Bellman's equation (2). For example, if we start with the true solution $V = \Gamma(V)$ and perturb it by adding a constant $C$ to $V$, the perturbed solution has a Bellman error of $(1-\beta)C$ which can be small even for large $C$ when $\beta$ is close to 1. Fortunately, DDCs do not require precise approximation of the level $V$ to accurately approximate the optimal policy in (3) since shifting $V$ by a constant leaves $\delta^*$ unaffected.[4]

### 2.4 Reinforcement Learning

These are stochastic iterative algorithms for approximating the value function $V$ and the optimal policy $\pi$ inspired by models of trial and error learning by animals and humans.[5] One of the first RL methods was *temporal difference learning* (TD) (Sutton, 1988) which can be regarded as a type of asychronous *stochastic approximation* (Robbins and Munro, 1951) for solving the linear system (4) for $V_\delta$ in the policy valuation step of PI. TD is a special case of *Q-learning* (Watkins, 1989) where $Q$ is the *choice-specific value function* implied by the Bellman equation

$$Q(s,a) = r(s,a) + \beta \sum_{s'} V(s')p(s'|s,a), \tag{6}$$

$$V(s) = \max_{a \in A(s)} Q(s,a). \tag{7}$$

$Q$-learning can be viewed as stochastic version of successive approximations $\{Q_t\}$ for solving (6) via the update rule

$$Q_{t+1}(s_t,a_t) = Q_t(s_t,a_t) + \alpha_t \left[ r(s_t,a_t) + \beta \max_{a \in A(s_t)} [Q_t(\tilde{s}_{t+1},a)] - Q_t(s_t,a_t) \right], \tag{8}$$

where $(s_t, a_t)$ is the state and action taken by the DM at time step $t$, and $\tilde{s}_{t+1}$ is a random draw from the transition probability $p(s'|s_t,a_t)$. Tsitsiklis (1994) showed that if the step size sequence $\{\alpha_t\}$ converges to 0 at a sufficiently slow rate, the sequence $\{Q_t\}$ converges with probability

[3] Though stochastic optimization methods such as stochastic gradient descent can be helpful in avoiding getting stuck at local minima, randomization does not break the curse of dimensionality of continuous non-convex optimization (Yudin and Nemirovsky, 1976).

[4] Fujimoto et al. (2022) show that Bellman error is a poor proxy for value error in offline RL. Nguyen (2025) shows that in DDC problems, we obtain more accurate solutions for $\delta^*$ by minimizing the sum of squared *anchored* Bellman errors, $V_\phi(s_j) - V_\phi(s_1) - [\Gamma(V_\phi)(s_j) - \Gamma(V_\phi)(s_1)]$ that difference out parallel shifts in $V_\phi$, allowing the minimizer to more accurately approximate the *shape* of $V$ by ignoring its level.

[5] See Sutton and Barto (2020) and Bertsekas (2025) for excellent introductions to this literature.

1 to the unique $Q$ that solves (6) and (7).[6] Equation (8) does not specify how actions $a_t$ is chosen: a typical choice is the *greedy policy* $a_t = argmax_{a \in A(s_t)} Q_t(s_t, a)$. A key condition for the convergence of Q-learning is that it samples each pair $(s, a)$ infinitely often, resulting in the usual exploration/exploitation tradeoff since suboptimal actions must be taken infinitely often for order for $Q$ learning to converge. But this will not happen under a greedy policy which is a pure strategy. One solution is to use a *$\varepsilon$-greedy policy* that takes the optimal action $a_t$ with probability $1 - \varepsilon$, and with probability $\varepsilon$ a randomly selected suboptimal action (i.e. an action $a$ with $Q_t(s_t, a) < Q_t(s_t, a_t)$). However unless $\varepsilon$ converges to 0 with $t$, $Q_t$ will converge to a $Q$ that does not satisfy the Bellman equation (7) since an $\varepsilon$-greedy policy is suboptimal.

Haarnoja et al. (2017), building on *maximum entropy reinforcement learning* (Ziebart et al., 2008), provided another solution to the exploration/exploitation tradeoff they call *soft-Q learning.* They modified the reward in the MDP to include an entropy term so the optimal policy is mixed rather than pure, guaranteeing that Q-learning results in continuous exploration with all feasible actions being chosen infinitely often, converging with probability 1 to the optimal mixed strategy $\pi$ given by the classic logit or softmax formula in equation (16) of section 3. The entropy of $\pi$ is $H(\pi(s)) = -\sum_{a \in A(s)} \pi(a|s) \log(\pi(a|s))$. Adding it to the reward results in the MDP

$$V(s) = \max_{\pi} V_\pi(s), \quad \text{where } V_\pi(s) = E_\pi \left\{ \sum_{t=0}^{\infty} \beta^t [r(s_t, a_t) + \sigma H(\pi(s_t))] | s_0 = s \right\}. \tag{9}$$

When the weight on entropy $\sigma = 0$ we obtain the original MDP objective (1) and the optimal $\pi$ is the pure strategy (3), but as $\sigma \to \infty$ the optimal policy entails choosing all feasible actions with equal probability, the maximum entropy solution. For $0 < \sigma < \infty$ the optimal $V$ solving (9) satisfies a "soft Bellman equation" identical to the "smoothed Bellman equation" derived by Rust (1988) for a "partially observed" MDP in equations (14) and (13) of section 3.

We say that RL is *model-free* if it does not require explicit specification or knowledge of $p(s'|s, a)$. Otherwise it is *model-based* RL. Q-learning can be *model-free,* unlike SA or PI, to the extent that it does not require a specification for the transition probability $p(s'|s, a)$ and avoids numerical integration to update $V$ and $Q$. This is possible when "the environment" generates new states rather than a computer simulation based on an explicit mathematical model for $p(s'|s, a)$. In Watkins' example of an animal learning to adapt and thrive, we do not need to know $p$ since the environment "draws" the successor states $s_{t+1}$ needed to implement the update in $Q_t$ in equation (8). When trained in this manner Q-learning is indeed model-free and so are other RL algorithms that interact directly with the environment (or use historical transitions for training/estimation as we discuss in section 4).

In many situations Q-learning and related RL methods produce policies that are too noisy and converge too slowly to be useful in real applications, especially when the number of states is large. As Jiang and Xie (2024) note, their "decisions can lead to undesirable outcomes, especially at the

[6] The conditions on the step size are 1) $\sum_t \alpha_t = \infty$ and 2) $\sum_t \alpha_t^2 < \infty$. Note $\alpha_t = 1/t$ satisfies these conditions.

early stages of learning when the algorithm has little knowledge of the environment. This is not a problem when the environment is a simulator, but can lead to serious consequences when people are part of the environment." As a result RL training is increasingly done in simulation environments with sufficient computer power to rapidly conduct many thousands or millions of training iterations to produce a much better solution than could be achieved from limited real-world experience. The ability to simulate from the true $p(s'|s,a)$ (or a good approximation to it) for a large number of iterations is key to producing good solutions using RL. For this reason most of the successful recent applications of RL are model-based.[7]

Tabular (i.e. finite state/action) Q-learning, (8), is far too slow for MDPs with large state or action spaces. A scalable solution to this problem to use a Deep Q-Network (DQN), which is a deep neural network with parameters $\phi$ that parameterize the Q-function as $Q_\phi$ similar to the way $V$ is approximated by $V_\phi$ in (5). This transforms the problem from iteratively updating Q-values, as in (8), to searching for $\phi$ that minimizes the mean squared Bellman error $\rho(\phi)$ given by

$$\rho(\phi) = \sum_{i=1}^{N} \left[ r(s_i, a_i) + \beta \sum_{s'} \max_{a' \in A(s_i)} Q_\phi(s', a') p(s'|s_i, a_i) - Q_\phi(s_i, a_i) \right]^2, \tag{10}$$

over a grid $\{(s_i, a_i)\}$. In many problems $\rho(\phi)$ be calculated because the transition probability $p$ is unknown, and, as already noted in section 2.3, minimizing Bellman error does not ensure that $Q_{\hat{\phi}}$ will closely approximate the true $Q$ and suboptimal policies can achieve low Bellman error due to overfitting (Fu et al., 2019).

Despite these challenges, Mnih et al. (2015) used a DQN to train effective strategies for classic Atari 2600 video games. Similar methods were subsequently used for board games such as chess, Go and Shogu, see Silver et al. (2018). They solved the problem of unknown $p$ by simulating game paths using *self-play* where each player makes choices using a trial policy $\pi$ resulting in a simulated data set $\mathcal{D}$ of $(s, a, s')$ sequences. Then they used stochastic gradient descent to minimize an approximation to the squared Bellman error (10) using the mean squared *temporal difference error*

$$\rho(\phi) = \mathbb{E}_{(s,a,s') \sim \mathcal{D}} \left[ \left( r(s,a) + \beta \max_{a'} Q_{\hat{\phi}}(s', a') - Q_\phi(s,a) \right)^2 \right], \tag{11}$$

where $\mathbb{E}_{(s,a,s') \sim \mathcal{D}}$ denotes the sample average over the simulated data set $\mathcal{D}$. Note that in (11) $\hat{\phi}$ is a lagged value of the network parameters from previous training iterations. They used $\hat{\phi}$ instead of $\phi$ in (11) due to the fact that the combination of off-policy learning, bootstrapping from subsequent estimates, and the use of a flexible nonlinear function approximator (the 'deadly triad') can lead to instability and non-convergence as noted by Sutton and Barto (1998).

[7] The use of Q-learning to train chess strategies is an interesting intermediate case. Although the functional form of $p(s'|s,a)$ for chess is unknown since this transition probability depends on the action of the opponent, Q-learning remains model-free since it relies on the environment to produce the next state $s_{t+1}$ directly via self-play, without explicitly modeling or using the game's rules to calculate $p(s'|s,a)$. In contrast, an algorithm like AlphaZero that uses Monte Carlo tree search explicitly uses the game rules for planning and would be considered model-based.

Two innovations in the DQN design avoided these problems. The first was *experience replay*, whereby transitions are stored in a large "replay buffer" $\mathcal{D}$. The network is then trained on mini-batches of transitions sampled uniformly randomly from $\mathcal{D}$, breaking the strong temporal correlations in the sequence of observations (e.g. persistent streaks in certain parts of the state space), thereby stabilizing the updates by conforming more closely to the i.i.d. assumptions underlying most stochastic gradient methods. The second innovation is the use of a separate *target network* with parameters $\hat{\phi}$, so that $Q_{\hat{\phi}}$ is a lagged version of the current network $Q_\phi$. Keeping the target network's parameters $\hat{\phi}$ fixed for several iterations of the overall stochastic descent algorithm, keeps the optimization target stable, preventing the current network from attempting to converge to a non-stationary objective. Together, these innovations mitigated the deadly triad and enabled Q-learning to scale to high-dimensional problems that were previously intractable.

## 3 Structural Estimation of DDCs

The term "structural" refers to the attempt to infer the underlying preferences and beliefs of economic agents rather than simply summarizing their behavior, the main goal of reduced-form econometrics. In a DDC model the agent's decision rule $\pi(a|s)$ is referred to as a "reduced-form model" that captures behavior: structural econometrics refers to methods that attempt to invert $\pi$ to infer the underlying preferences and beliefs $\{\beta, r, p\}$ that generate observed behavior. Key references include Rust (1987), Rust (1994) and the survey by Aguirregabiria and Mira (2010).[8]

An obstacle to inference is the fact that the optimal policy $\delta^*$ in (3) is a pure strategy (i.e. deterministic function of $s$). This leads to a problem of *statistical degeneracy* because different individuals in the same observed state $s$ will typically make different choices, something the model cannot explain. To explain this heterogeneity we augment the state variable as $x = (s, \varepsilon)$ where $\varepsilon$ is a vector of idiosyncratic preference shocks observed by the agent but not by others. Even though $\delta^*(x, \varepsilon)$ is still a pure strategy from the agent's standpoint, it produces choices that appear random from the observer's standpoint. Under additional assumptions that include the assumption $A(s)$ is finite for all $s \in S$, we can derive a *conditional choice probability* (CCP) $\pi(a|s)$ implied by the strategy $\delta^*$ that admits positive probabilities for all feasible choices $a \in A(s)$. Using this, we can form a likelihood function to infer $\{\beta, r, p\}$ from data on agents' states and choices.

Rust (1987) extended McFadden (1973)'s work on static random utility models of discrete choice to DDC models, resulting in an analytic formula for the CCP $\pi(a|s)$ known as the *multinomial logit model* ("softmax" function in the RL literature). Suppose that for each $s$, $\varepsilon$ is a vector with the same length as $|A(s)|$ the number of actions, so for $\varepsilon(a)$ is the component of $\varepsilon$ associated with action

[8] See Rust (2014) for a discussion of the origins of SE, dating back to work at the Cowles Foundation in the 1940s and the rationale for this literature including the argument of Robert E. Lucas (1976) that structural models are superior to reduced-form for counterfactual prediction of policy changes. The development of DP in the 1950s and 60s revolutionized economics by enabling models of sequential decision making under uncertainty, dynamic strategic interactions, and dynamic equilibria in markets which are key features of nearly all economic problems. DP lead to new dynamic economic theories based on *rational expectations* that showed how beliefs about uncertain future events affect the current choices of rational forward looking agents.

$a \in A(s)$. Assume that the reward for action $a$ in state $(s, \varepsilon)$ can be written as $r(s, a) + \varepsilon(a)$. We also assume that the transition density $p(s', \varepsilon'|s, \varepsilon, a)$ factors as $g(\varepsilon'|s')p(s'|s, a)$, for a conditional density $g(\varepsilon'|s')$, where $\varepsilon$ has a multivariate Gumbel or Type 1 extreme value (EV) distribution with common scale parameter and expectation equal to $E\{\varepsilon(a)|s\} = 0$ for $a \in A(s)$. Let $V(s, \varepsilon)$ be the optimal value of this MDP, the solution to the Bellman equation (2). Under these assumptions we can show $V(s, \varepsilon)$ has the representation

$$V(s, \varepsilon) = \max_{a \in A(s)} [Q(s, a) + \epsilon(a)], \tag{12}$$

where $Q = \Lambda_\sigma(Q)$ is the unique fixed point to the "soft Bellman operator" $\Lambda_\sigma$ given by

$$\Lambda_\sigma(Q)(s, a) = r(s, a) + \beta \sum_{s'} \sigma \log \left( \sum_{a' \in A(s')} \exp\{Q(s', a')/\sigma\} \right) p(s'|s, a). \tag{13}$$

The derivation of (12) and (13) use an important property of the EV distribution for $\varepsilon$, namely that the expectation of $V(s, \varepsilon)$ with respect to $\varepsilon$ has an analytical expression given by the so-called log-sum or "soft max function" $V(s)$

$$V(s) \equiv E\left\{V(\tilde{s}, \tilde{\varepsilon}) \,|\tilde{s} = s\right\} = \int_\varepsilon \max_{a \in A(s)} [Q(s, a) + \varepsilon(a)] \, g(d\varepsilon|s) = \sigma \log \left( \sum_{a \in A(s)} \exp\{Q(s, a)/\sigma\} \right). \tag{14}$$

Using equations (13) and (14) we can derive a "soft Bellman equation" for $V$

$$V(s) = \sigma \log \left( \sum_{a \in A(s)} \exp \left\{ [r(s, a) + \beta \sum_{s'} V(s') p(s'|s, a)]/\sigma \right\} \right). \tag{15}$$

Equation (13) defines a "soft $Q$" as the fixed point of the smoothed Bellman operator $Q_\sigma = \Lambda_\sigma(Q_\sigma)$, and (15) defines $V$ as a fixed point of the smoothed Bellman operator $V_\sigma = \Gamma_\sigma(V_\sigma)$ and both are contractions. These operators are equivalent to the corresponding operators for MDP without the state variable $\varepsilon$ in equations (6) and (7), and it is not hard to show that $Q_\sigma$ and $V_\sigma$ converge to $Q$ and $V$ as $\sigma \downarrow 0$.

The representation of $V(x, \varepsilon)$ in equation (12) means that the decision rule $\pi(a|x, \varepsilon)$ is isomorphic to a static discrete choice model where $Q(s, a) + \varepsilon(a)$ is the utility/payoff from action $a$. The CCP $\pi(a|s)$ is the conditional probability that the DM chooses action $a \in A(s)$ given the observed state $s$, taking the multinomial logit form

$$\begin{aligned} \pi(a|s) &= \int I \left\{ Q(s, a) + \varepsilon(a) \geq \max_{a' \in A(s)} [Q(s, a') + \varepsilon(a')] \right\} F(d\varepsilon|s) \\ &= \frac{\exp\{Q(s, a)/\sigma\}}{\sum_{a' \in A(s)} \exp\{Q(s, a')/\sigma\}}. \end{aligned} \tag{16}$$

As $\sigma \downarrow 0$, $\pi(a|s)$ converges to a degenerate probability that equals 1 if $Q(s,a)$ is the largest $Q$ value and 0 otherwise, i.e. to the optimal pure strategy $\pi$ in equation (3) for MDPs without the $\varepsilon$ variable. If follows that the choice-specific values $Q(s,a)$ in (13) and CCPs $\pi(a|s)$ in (16) are exactly the same as the "soft Q" values and optimal policy derived in the RL literature under the assumption that agents care about entropy of the policy $\pi$, given in equation (9) of section 2.4.

Aguirregabiria and Mira (2002) showed how to implement a "soft" analog of *policy iteration,* see section 2.2, to solve the soft Bellman equation (15). Equation (16) is the *policy improvement step,* the analog of (3), that computes the policy $\pi$ implied by $Q$. The analog of the *policy valuation* step (4) is the function $V_\pi$ that solves the linear system

$$V_\pi(s) = \sum_{a \in A(s)} \pi(a|s)[r(s,a) + E_\pi\{\varepsilon(a)|s,a\} + \beta \sum_{s'} V_\pi(s')p(s'|s,a)], \tag{17}$$

where $E_\pi\{\varepsilon(a)|s,a\}$ is the conditional expectation of $\varepsilon(a)$ given that action $a$ is chosen in state $s$ under decision rule $\pi$. The EV assumption implies that $E_\pi\{\varepsilon(a)|s,a\} = -\sigma \log(\pi(a|s))$, and using equation (16), we see that right side of (17) equals the soft max function defining the soft Bellman operator in equation (15). Soft policy iteration solves the soft Bellman equation $V_\sigma = \Gamma_\sigma(V_\sigma)$ by cycling between policy valuation steps (17) and policy improvement steps (16), using the $Q_\pi$ implied by $V_\pi$ in equation (13). Hard/soft policy iteration is equivalent to using Newton's method to solve the hard/soft Bellman equations for $V$.

Equation (16) implies that $\sigma \log(\pi(a|s)) = Q(s,a) - V(s) \equiv \mathcal{A}(s,a)$, where $\mathcal{A}(s,a)$ is the advantage function. We can use it to rewrite the fixed point equation (13) for $Q$ as a linear system depending on $r(s,a)$ and $\log(\pi(a|s))$,

$$Q(s,a) = r(s,a) + \beta \sum_{s'} \sum_{a' \in A(s')} \left[-\sigma \log(\pi(a'|s')) + Q(s',a')\right] \pi(a'|s')p(s'|s,a). \tag{18}$$

Thus, $Q$ decomposes as $Q(s,a) = Q_r(s,a) + Q_\varepsilon(s,a)$ where $Q_r(s,a)$ is the component of discounted value from observed rewards $r$ and $Q_\varepsilon(s,a)$ is the component from unobserved rewards $\varepsilon$. Further substituting expression (17) in place of (15) in the equation for $Q$ in (13), we can show that $Q_r$ and $Q_\varepsilon$ are each the solution to the following linear systems

$$Q_r(s,a) = r(s,a) + \beta \sum_{s'} \sum_{a' \in A(s')} Q_r(s',a')\pi(a'|s')p(s'|s,a) \tag{19}$$

$$Q_\varepsilon(s,a) = \beta \sum_{s'} \sum_{a' \in A(s')} \left[-\sigma \log(\pi(a'|s')) + Q_\varepsilon(s',a')\right] \pi(a'|s')p(s'|s,a), \tag{20}$$

a decomposition that reduces the computational burden of estimating DDCs.

### 3.1 Estimation Methods for DDCs

The most common estimation method for DDCs is some form of maximum likelihood. Suppose we observe a random sample of *trajectories* i.e. data on the consecutive states and actions taken by $N$ individuals $\{(s_{it}, a_{it})|t = 0, \ldots, T_i,\ i = 1, \ldots, N\}$. Suppose that we smoothly parametrize $(r, p)$ using a vector unknown parameters $\theta$, which we denote by $r_\theta$ and $p_\theta$.[9] Using the implicit function theorem, we can show that $V_\theta$ and $Q_\theta$, the contraction fixed points given in equations (14) and (13), are continuously differentiable function of $\theta$. Using these fixed points and the logit formula for $\pi(a|s)$ in equation (16) we can write a full likelihood function for the data as a function of the unknown parameters $\theta$ given by

$$L(\theta) = \prod_{i=1}^{N} \prod_{t=1}^{T_i} \pi_\theta(a_{it}|s_{it}) p_\theta(s_{it}|s_{it-1}, a_{it-1}). \tag{21}$$

The smoothness of $V_\theta$ and $Q_\theta$ in $\theta$ implies that the CCP $\pi_\theta$ and the likelihood $L(\theta)$ are both smooth functions of $\theta$ so the standard asymptotic efficiency properties of maximum likelihood apply.

The Nested Fixed Point algorithm (NFXP) (Rust, 1988) computes the MLE of $\theta$ via a standard outer hill-climbing loop that maximizes $L(\theta)$, where each evaluation calculates the contraction fixed point $Q_\theta = \Lambda_\theta(Q_\theta)$ that defines the choice-specific values $Q_\theta$ where $\Lambda_\theta$ is the operator defined on the right hand side of equation (13). Notice that $L(\theta)$ depends on $\theta$ via $Q_\theta$. An alternative strategy (Su and Judd, 2012) computes the MLE via the MPEC algorithm, maximizing $L(\theta, Q)$ as a function of $(\theta, Q)$ subject to the fixed point constraint $Q_\theta = \Lambda_\theta(Q_\theta)$.[10] NFXP's computational burden stems from computing the inner fixed point $Q_\theta = \Lambda_\theta(Q_\theta)$ for each trial value of $\theta$ in the outer hill-climbing algorithm. Newton's method (e.g. policy iteration) is typically used to solve for $Q_\theta$ or for $V_\theta$ from the soft Bellman equation (15). Even though Newton's method converges rapidly, it requires $O(|S|^3)$ operations per Newton/policy iteration step, making NFXP burdensome for large state spaces $S$. Therefore alternative estimation methods have been proposed that reduce the total number of policy iteration steps required to estimate $\theta$.

The "CCP estimator" (Hotz and Miller, 1993) for $r$ uses a non-parametric estimate of $\pi(a|s)$ to avoid repeated solution of $Q$ over the search for $\theta$ that NFXP requires. Consider the case where $r_\theta$ is linear in parameters, i.e. we can write $r_\theta(s, a) = r(s, a) * \theta \equiv \sum_{k=1}^{K} r_k(s, a)\theta_k$ for known functions $\vec{r} = (r_1, \ldots, r_K)$ called *features* in the IRL literature. Then (19) implies that $Q_r(s, a) = R(s, a) * \theta$ where $R$ is the solution to

$$R(s, a) = \vec{r}(s, a) + \beta \sum_{s'} \sum_{a' \in A(s')} R(s', a')\pi(a'|s')p(s'|s, a) = \vec{r}(s, a) + \beta ER(s, a), \tag{22}$$

[9] The discount factor $\beta$ is often treated as known, though in some cases it is also estimated and thus part of $\theta$.

[10] Su and Judd (2012) claim that the MPEC alogithm is substantially faster than NFXP, but this was due to using SA (Iskhakov et al., 2016) to compute the fixed point $Q_\theta = \Lambda_\theta(Q_\theta)$, which is slow when $\beta$ is close to 1. When Newton's method is used to compute this fixed point, the cpu times for MPEC and NFXP are roughly comparable with NFXP being faster on test problems with large sample sizes.

where $ER(s,a)$ is a shorthand for the conditional expectation of $R(s,a)$ the middle of expression in (22). It follows that we only need to solve (22) once using a non-parametric estimate of $\hat{\pi}$ along with $Q_\varepsilon$ in (20) to serve as a "correction term" to obtain the following form for the CCP

$$\pi_\theta(a|s) = \frac{\exp\{\hat{R}(s,a)\cdot\theta + \hat{Q}_\varepsilon(s,a)\}}{\sum_{a'\in A(s)}\exp\{\hat{R}(s,a')\cdot\theta + \hat{Q}_\varepsilon(s,a')\}}, \tag{23}$$

where $\hat{R}(s,a)$ is the solution to (22) and $\hat{Q}_\varepsilon$ is the solution to (20) using non-parametric first stage estimators $\hat{\pi}(a|s)$ and $\hat{p}(s'|s,a)$. The CCP estimator $\hat{\theta}$ maximizes the partial likelihood $L_\pi(\theta)$ given by

$$L_\pi(\theta) = \prod_{i=1}^{N}\prod_{t=1}^{T_i}\pi_\theta(a_{it}|s_{it}), \tag{24}$$

where $\pi$ is given in (23).[11] Thus, when $r$ is a linear combination of known features, we only need to compute $\hat{R}$ and $\hat{Q}_\varepsilon$ once at the start of the estimation, requiring $K+1$ solutions of systems with $|S|$ equations and unknowns, where $K$ is the number of features. This can be further reduced to a single solution if we assume that $r(s,a_s)$ is known for some "normalizing alternative" $a_s \in A(s)$ (see Section 3.3).

A cost of the CCP estimator is that it is less efficient than using NFXP to estimate the same partial likelihood criterion (24) due to the noise in the first stage non-paramteric estimates $\hat{\pi}$ that are used to construct the $\hat{Q}_\varepsilon$ function in (23). Aguirregabiria and Mira (2002) addressed this problem by proposing an iterative estimator called *nested pseudo likelihood* (NPL) that uses an initial non-parametric estimate $\hat{\pi}$ to compute the value function $V_{\hat{\pi}}$ implied by this initial estimate by solving for it in the policy valuation step in equation (17), which we denote as $\hat{V}_{\theta,\hat{\pi}}$ since it depends both on $\hat{\pi}$ and the parameters $\theta$ entering the reward function. NPL estimates $\hat{\theta}$ using a pseudo-likelihood similar to (24) except that the CCP $\pi_\theta(a|s)$ is evaluated using (16) using estimated $Q$ functions given by

$$\hat{Q}_\theta(s,a) = r_\theta(s,a) + \beta\sum_{s'}\hat{V}_{\theta,\hat{\pi}}(s'|s,a)p(s'|s,a), \tag{25}$$

Thus, the likelihood $L_\pi(\theta)$ as depends implicitly on the first stage non-parametric CCP estimate $\hat{\pi}$, which can be regarded as a "nuisance parameter". In principle, estimation noise in $\hat{\pi}$ will contaminate and affect the asymptotic covariance matrix for the parameters of interest $\theta$. However at the fixed point, we have $\partial V(s)/\partial\pi = 0$ the so-called *zero Jacobian property* that implies that the first stage nuisance parameter $\hat{\pi}$ is "Neyman orthogonal" with respect to the parameters of interest $\theta$ in the sense of Chernozhukov et al. (2022). It follows that the asymptotic covariance matrix of the NPL estimator of $\theta$ is unaffected by noise in the first stage $\hat{\pi}$. NPL can be iterated, using $\hat{\theta}$ to update the estimate of the CCPs using equation (16), then using the updated policy $\hat{\pi} = \pi_{\hat{\theta}})$ and a soft policy valuation step (18) to obtain updated $V$ and $Q$ functions (25), and use these

[11] See Chernozhukov et al. (2022) for a modified version of the CCP estimator that is "locally robust" i.e. it corrects for sampling error in the first stage estimates of the functions $\hat{H}(s,a)$ and $Q_\varepsilon(s,a)$.

in the likelihood (24) to get an updated estimate $\hat{\theta}$. Iterative NPL amounts to "swapping" the policy iteration and maximization steps of the NFXP estimator (NFXP maximizes the likelihood over $\theta$ in the "outside" loop and does policy iteration in the "inside" loop). Swapping the order of maximization and policy iteration reduces the total number of policy iteration steps required to estimate $\theta$. Aguirregabiria and Mira (2002) showed that if the iterated version of NPL converges, it produces the same estimate $\hat{\theta}$ that NFXP produces from maximization of the partial likelihood (24). Thus the iterated NPL estimator is as efficient but faster than NFXP and more efficient but slower than the CCP estimator.[12]

The extreme value (EV) assumption, while restrictive, dramatically reduces computational burden via the analytic expressions it implies for the softmax CCP (16) and expected value function (14). Other natural choices for the unobserved $\varepsilon$ such as multivariate normal do not result in analytical expressions and thus considerable computation to approximate these quantities. The conditional independence (CI) of the $\{\varepsilon_t\}$ shocks is another restrictive assumption driven by computational considerations. Though there are computationally efficient recursive likelihood integration methods, e.g. Reich (2018), that can allow for serial correlation in $\varepsilon_t$ shocks, these methods are still burdensome for estimation. Imai et al. (2009) and Norets (2009) introduced Bayesian inference in DDC models that use Markov Chain Monte Carlo (MCMC) methods to simulate from rather than directly compute the posterior by numerical integration. MCMC facilitates *data augmentation* where latent variables can be simulated symmetrically with observed ones rather than being marginalized from the likelihood, allowing us to relax the EV and CI assumptions. They also show that RL can be used to update the Q functions of the underlying MPD problem in parallel inside the overall do-loop for the MCMC simulations where the structural parameters are drawn from the likelihood using Gibbs' or Metropolis-Hastings' samplers.

There are also non-Bayesian, non-likelihood based simulation estimators such as the *method of simulated moments* (MSM) McFadden (1989), *indirect inference* Gourieroux et al. (1993), the *moment inequality* estimator of Bajari et al. (2007), and the *adversarial estimator* of Kaji et al. (2023) that use stochastic simulations to estimate problems where it is difficult to explicitly specify or compute a likelihood function. For example, Hall and Rust (2021) apply MSM to structurally estimate a dynamic model of inventory investment in steel plates, where a key state variable, the price of steel plate, is only recorded by the company only on the dates it buys steel plates. Since the company only occasionally purchases new inventory, maximum likelihood estimation entails high dimensional integration over the unobserved price of steel on the majority of days the firm does not buy steel plate. With MSM we can simulate prices regardless of whether the firm buys or not, and MSM searches for $\theta$ to mimimize the distance between moments we construct from observed data to equivalent moments from simulations of the structural model.

[12]See also *efficient pseudo likelihood* (EPL) (Dearing and Blevins, 2025), a similar iterative procedure that is also an efficient sequential estimator that can estimate parameters of dynamic games.

### 3.2 Function Approximation Methods for Large State Spaces

When the state space $S$ is large (i.e. the DDC has many continuous state variables that must be discretized) the approaches discussed above that use PI and repeated solution of large linear systems with $|S|$ equations and unknowns are computationally infeasible. Following section 2.3 we can approximate the fixed point $Q = \Lambda_\sigma(Q)$ of the soft Bellman operator in equation (13) by $Q_\phi(s)$ with a flexible parametric family such as a neural network with weights $\phi$, and solve the fixed point problems (13) and (14) more tractably as nonlinear least squares problems. Let $\rho(\theta, \phi)$ be the sum of squared residuals of the soft Bellman operator, $Q_\phi - \Lambda_\sigma(Q_\phi)$, over a grid of $N$ points in the state space, equation (10). Note that the residuals also depend on $\theta$ via $\{\beta, r, p\}$ in the soft Bellman operator $\Lambda_\sigma$. A nested least squares estimator similar to NFXP can estimate $\theta$ by maximizing the likelihood $L(\theta)$ (21) while mimimizing $\rho(\theta, \phi)$ over $\phi$ for each value of $\theta$. A penalized likelihood estimator called SEES (Luo and Sang, 2024), for *sieve-based efficient estimator for structural models*, chooses $\hat{\theta}$ and $\hat{\phi}$ to maximize

$$(\hat{\theta}, \hat{\phi}) = \underset{\theta,\phi}{argmax} \log(L(\theta)) - \lambda\rho(\theta, \phi), \tag{26}$$

where $\lambda \geq 0$ is a penalization parameter. Consistency of SEES requires the use of a *sieve,* i.e. the dimension of $\phi$ must grow to infinity at certain rate with the sample size so that the approximation error in approximating $Q$ by $Q_\phi$ converges to zero. A related estimator, the *neural network efficient estimator* (NNES) of Nguyen (2025), maximizes a penalized criterion similar to SEES in a modified version of the sequential NPL estimator of Aguirregabiria and Mira (2002).

A challenge facing sieve-based estimators is that $\phi$ are high dimensional "nuisance parameters" (their only purpose is to approximate $Q$) whereas $\theta$ are the "parameters of interest" that determine rewards and beliefs, $\{\beta, r, p\}$. Since $\hat{\phi}$ minimizes the sum of squared residuals $\rho(\phi, \theta)$ for each $\theta$, it follows the $\hat{\phi}$ will be an implicit function of $\theta$. The asymptotic covariance matrix for $\hat{\theta}$ requires $\nabla_\theta Q$, the gradient of $Q$ with respect to $\theta$. Diifferentiating $Q_{\hat{\phi}(\theta)}$ with respect to $\theta$ using the first order condition for $\hat{\phi}(\theta)$, $\nabla_\phi Q_\phi = 0$ at $\phi = \hat{\phi}(\theta)$, requires the Hessian $\nabla^2_{\phi,\phi} Q_\phi$ which is typically infeasible to compute. Thus computing $\nabla_\theta Q = \nabla_\phi Q_{\hat{\phi}(\theta)} \nabla_\theta \hat{\phi}(\theta)$ via the chain rule is out of the question. Computing $\nabla_\theta Q$ via numerical differentiation of $Q_{\hat{\phi}(\theta)}$ with respect to $\theta$ is also time-consuming and potentially inaccurate because multiple local optima in $\phi$ result in fragile solutions $\hat{\phi}(\theta)$ where small changes in $\theta$ produce discontinuous jumps in $\hat{\phi}(\theta)$ and $Q_{\hat{\phi}(\theta)}$. Nguyen (2025) shows that a fast and accurate way to compute $\nabla_\theta Q$ is to use the method of *equilibrium propagation* introduced by Zucchet and Sacramento (2022).

### 3.3 The Identification Problem

Section 2 showed how $\pi$ is calculated from the structure $\{\beta, r, p\}$, which we summarize by the mapping $\pi = f(\beta, r, p)$. The Identification Problem asks if this mapping is invertible: given $\pi$ is there a unique underlying structure $\{\beta, r, p\}$ that implies it? Unfortunately, without further

restrictions the answer is negative, as shown by Hotz and Miller (1993), Rust (1994), Ng et al. (1999), and Magnac and Thesmar (2002). Consider the first step of the inversion process, uncovering $Q$ from $\pi$:

$$\log\big(\pi(a|s)/\pi(a'|s)\big) = Q(s,a) - Q(s,a'), \quad s \in S,\ a, a' \in A. \tag{27}$$

Thus, knowledge of $\pi$ is insufficient to recover all the $Q$ values: only the *differences* in $Q$ are identified. Indeed, the structure $\{\beta, r, p\}$ contains $1 + |S|(|A|-1) + |S|^2|A|$ parameters, more than then $|S|(|A|-1)$ parameters of $\pi$, so we will generally have "more equations than unknowns" so the structure is only partially identified. Unique (or "point") identification requires additional restrictions on $\{\beta, r, p\}$. A common assumption is that $r$ and $p$ have known parametric functional forms that depend on a relatively small number of parameters $\theta$. In some situations the reward function can be treated as known, and this facilitates the identification of subjective beliefs $p(s'|s,a)$.[13] Rational expectations is a strong identifying assumption, i.e. agents' subjective beliefs $p(s'|s,a)$ correspond to the actual probability governing these transitions. With sufficient data, $p$ can be estimated nonparametrically and treated as known from the standpoint of identification. The discount factor $\beta$ is often assumed to be known *a priori,* though Abbring and Daljord (2020) showed that $\beta$ is identified under rational expectations with certain restrictions on rewards. In the IRL setting, Yao et al. (2024) take a similar approach, demonstrating that observing multiple experts who share a reward function but operate under different planning horizons allows for joint identification of both the reward structure and discount factors.

When $\{\beta, p\}$ are known, identification of $r$ depends on the assumption that either it has a known parametric functional form $r_\theta$ where the dimension of $\theta$ is less than $|S|(|A|-1)$, or that for each $s \in S$ there is a normalizing or "anchor action" $a_s$ such that $r(s, a_s)$ is known. The latter assumption amounts to $|S|$ restrictions that imply there are only $|S|(|A|-1)$ unknown rewards to be recovered, the same number of free values of $\pi$. Hotz and Miller (1993) and Kang et al. (2025) show that in this "exactly identified" case, we can calculate $Q(s, a_s)$ from knowledge of $r(s, a_s)$ and $\pi(a_s|s)$ using equation (18) and then (27) allows us to recover all remaining $Q$ values. Using the $Q$ values we can back out the remaining unknown rewards $r$ from the soft Bellman equation (13).[14]

The assumption that $r(s, a_s)$ is known for some normalizing action $a_s$ in each state $s \in S$ is far from innocuous. If wrong, estimated rewards will not equal true rewards even though the model perfectly fits $\pi$, and counterfactual predictions may differ from actual outcomes. For example we can always estimate rewards with the trivial estimator $\hat{r}(s,a) = \log(\pi(a|s))$ that perfectly fits observed behavior and is optimal, i.e. the implied $Q$ function satisfies the fixed point $Q = \Lambda_\sigma(Q)$ in (13). However easy to see this estimator differs from the true reward: $\hat{r}(s,a) = r(s,a) +$

[13] For example Anderson et al. (2025) are able to identify the subjective beliefs of professional tennis servers about how their serve strategies affects their probability of winning a tennis game. These beliefs take the form of a transition $p(s'|s,a)$ where $s$ denotes the point state of the game and $a$ denotes serve direction. Identification of beliefs is possible because it is reasonable to assume $r$ is known: i.e. the server earns a reward of 1 if they win the game and 0 otherwise, and due to reasonable parametric restrictions on $p$.

[14] van der Laan et al. (2026) provid an alternative condition that implies exact identification or $r$, namely that there exists some known policy $\pi(a|s)$ with zero expected reward, $0 = \sum_{a \in A(s)} r(s,a)\pi(a|s)$, for all $s \in S$.

$\beta \int V(s')p(s'|s,a) - V(s)$. Indeed, in the absence of any prior restrictions on rewards, we can only identify a set of rewards $\mathcal{R}(r) = \{r_h\}$ that are observationally equivalent to $r$ given by $r_h(s,a) = r(s,a) + \beta \sum_{s'} h(s')p(s'|s,a) - h(s)$ for any function $h : S \to R$. Ng et al. (1999) refers to $\mathcal{R}(r)$ as "potential-based reward shaping" and shows that *policy invariance* holds: i.e. $\pi_{r_h} = f(\beta, r_h, p) = \pi_r = f(\beta, r, p)$ for all $r_h \in \mathcal{R}(r)$. However for alternative transition probabilities $\lambda(s'|s,a) \neq p(s'|s,a)$ policy invariance does not hold: $\pi_{r_h} = f(\beta, r_h, \lambda) \neq \pi_r = f(\beta, r, \lambda)$, i.e. counterfactual behavior implied by observationally equivalent rewards $r_h$ and $r$ differ in the new environment $\lambda$. While this sensitivity complicates counterfactuals, it can be leveraged for identification: Cao et al. (2021) prove that rewards are identifiable up to a constant if the agent is observed in at least two environments with sufficiently different transition laws, effectively resolving the entanglement between dynamics and rewards.

The DDC and IRL literatures use different assumptions to overcome the identification problem, reflecting different goals of the respective fields. van der Laan et al. (2026) and Kalouptsidi et al. (2021) show that some counterfactuals are identified even if $\{\beta, r, p\}$ is partially identified, providing a "call for caution while leaving room for optimism: although counterfactual behavior and welfare can be sensitive to identifying restrictions imposed on the model, there exists important classes of counterfactuals that are robust to such restrictions." (p. 385). Similarly, Rolland et al. (2022) prove that full parameter identification is unnecessary for transfer; observing experts in environments with sufficiently linearly independent transition matrices satisfies a rank condition that guarantees the learned reward will induce the optimal policy in a new environment, even if the reward itself is not unique.

## 4 Inverse Reinforcement Learning

The IRL literature (Ng and Russell, 2000) has two motivations for inferring rewards: 1) "the potential use of reinforcement learning and related methods as potential models for human and animal learning" and 2) to provide a reward function that RL can use to construct an "intelligent agent that can behave successfully in a particular domain." Though they note that imitation and apprenticeship learning (IL and AL) can be used to learn a policy $\pi$, "the reward function often provides a much more parsimonius description of behaviour" so that IRL can constitute an effective form of AL. IRL has become an important method in many areas of robotics and Wang et al. (2025) note that IRL has "found great success in predicting trajectories through inferring cost or reward functions from expert demonstrations, and then using these functions to guide the behavior of self-driving vehicles in unseen driving environments." The idea of inferring objectives from observed optimal behavior has a longer history, tracing to Kalman (1964)'s work on Inverse Optimal Control in engineering.

The IRL problem (Ng and Russell, 2000) is to infer the reward function $r_\theta(s,a) = \theta^\top \phi(s,a)$ from expert demonstrations. Given trajectories $\mathcal{D}_E = \{\tau_i\}$ or a policy $\pi_E$, we seek the parameters $\theta$ that rationalize observed behavior. The problem is ill-posed: many $\theta$ can yield the same observed

behavior. For linear rewards, Abbeel and Ng (2004) showed that matching the expert's feature expectations $\mu(\pi) = \mu_E$ suffices for policy imitation but does not uniquely identify $\theta$. Different IRL algorithms represent different principles for selecting a unique $\theta$ from this equivalence class. We selectively review methods most relevant to DDC and economic applications, referring readers to Azar et al. (2020), Arora and Doshi (2021), Gleave and Toyer (2022), and Adams et al. (2022) for comprehensive surveys.

While feature matching suffices for policy imitation, it does not uniquely identify rewards. Maximum margin methods (Ratliff et al., 2006) address this by requiring the expert's cumulative value to exceed any alternative policy's value by a margin that grows with the alternative's deviation from expert behavior. Policies vastly different from the expert's are penalized more heavily. This framework resembles moment inequality estimators in dynamic games (Bajari et al., 2007). However, margin-based methods assume a near-optimal expert, making them brittle in stochastic environments with noisy demonstrations. Furthermore, the constraint set is exponentially large, requiring computationally intensive methods. These limitations motivated probabilistic approaches that explicitly model suboptimal behavior.

### 4.1 Maximum Entropy and Maximum Likelihood Methods

Chronologically, the first use of a Boltzmann likelihood in IRL was Ramachandran and Amir (2007), who modeled expert actions as $P(a|s, R) \propto \exp(\alpha Q^*(s, a, R))$ where $\alpha$ is a "confidence parameter" equivalent to the inverse temperature $1/\sigma$ in DDC.[15] Crucially, BIRL uses standard ("hard") Q-values from the classical Bellman equation, treating stochastic choice as optimization error. They combine this likelihood with a prior on rewards and use MCMC to estimate the posterior $P(R|\mathcal{D}_E)$.

Building on this foundation, subsequent work derived the same likelihood from different principles. Modern IRL methods rely on a softmax (Boltzmann-rational) policy as their probabilistic foundation:

$$\pi_\theta(a|s) = \frac{\exp(Q_\theta(s, a)/\sigma)}{\sum_{a'} \exp(Q_\theta(s, a')/\sigma)} \tag{28}$$

This is identical to the CCP formula (16) from DDC.[16] There are a few ways to motivate such a soft-optimal policy.

Ziebart et al. (2008, 2010) derive (28) by maximizing causal entropy $H_c(\pi) = \mathbb{E}_\pi[-\sum_t \log \pi(a_t|s_t)]$ subject to matching the expert's feature expectations: $\mathbb{E}_\pi[\phi] = \mu_E$.[17] The reward parameters $\theta$ emerge as Lagrange multipliers. This information-theoretic derivation produces the most random

[15] Bayesian IRL (2007) predates the Bayesian DDC literature (Imai et al., 2009; Norets, 2009) by two years, yet neither community cited the other, an instance of parallel discovery across disciplinary silos.

[16] Small $\sigma$ makes the policy concentrate almost entirely on the highest-value action, while large $\sigma$ spreads probability nearly uniformly across actions. Because the softmax formula depends only on $Q/\sigma$, the reward scale and $\sigma$ cannot both be identified, so DDC applications typically fix $\sigma = 1$. When calibration data are available, Ghosal et al. (2023) show how to estimate the inverse temperature $\alpha$ (equivalently $\sigma = 1/\alpha$) instead of fixing it.

[17] Causal entropy differs from Shannon entropy over trajectories: it conditions each action only on past and current states, respecting the MDP's temporal structure. Standard Shannon entropy $H(\tau) = -\sum_\tau P(\tau) \log P(\tau)$ does not impose this causal constraint.

policy consistent with observed behavior. This MaxEnt approach is agnostic about why behavior appears stochastic: it does not assume agents optimize-but-tremble (as in BIRL) or face unobserved utility shocks (as in RUM). The principle traces to Jaynes (1957), who showed that maximizing entropy subject to moment constraints yields the least presumptive distribution consistent with the data. The solution assigns trajectory probabilities proportional to exponentiated cumulative rewards, $p(\tau) \propto \exp(\sum_t \gamma^t r(s_t, a_t)/\sigma)$, so higher-reward trajectories are exponentially more likely. The Lagrangian dual of this entropy maximization problem is equivalent to maximum likelihood estimation of $\theta$ from the observed trajectories, establishing a deep connection between information-theoretic and statistical perspectives.

A second motivation comes from Random Utility Theory. Pitombeira-Neto et al. (2024) argue the agent is not stochastic but deterministically optimal; the observed randomness arises from the observer's incomplete information about utility-relevant features. They model total reward as $R(s, a) = r_\theta(s, a) + \epsilon$, where $\epsilon$ captures unobserved features known only to the agent. If $\epsilon$ follows a Gumbel distribution, the resulting choice probability is identical to (28). This provides a microfoundation rooted in incomplete information rather than bounded rationality, and they prove Maximum Entropy IRL is a special case of their Random Utility IRL framework with $\gamma = 1$. Matejka and McKay (2015) provide yet another microfoundation: under Rational Inattention, agents optimally allocate limited attention, and the resulting choice probabilities take the logit form.

A third derivation makes use of a behavioural assumption. Assume experts choose actions according to (28). MLE of $\theta$ from demonstrations $\mathcal{D}_E$ yields the gradient:

$$\nabla_\theta \mathcal{L}(\theta) \propto \mu_E - \mathbb{E}_{\pi_\theta}[\phi] \tag{29}$$

At the optimum, model feature expectations match expert feature expectations, the same condition as the Max-Ent constraint. The partial likelihood is identical to (24) from DDC, establishing mathematical equivalence between the two literatures under specific conditions: linear rewards and an exponential family likelihood (Berger et al., 1996). While Ziebart et al. (2010) proves this equivalence (Thm 6.4), the MaxEnt framework provides a distinct motivation rooted in handling ambiguity rather than explicit agent optimization, remaining useful when RUM assumptions of DDC may not hold.[18] Max-Ent IRL optimizes over trajectory distributions rather than action-by-action choices, accounting for temporal correlations at the cost of increased computational complexity.

The Max-Ent IRL algorithm iterates until convergence. First, a backward pass computes soft value functions via $Q_\theta(s, a) = r_\theta(s, a) + \beta \sum_{s'} p(s'|s, a) V_\theta(s')$ and $V_\theta(s) = \sigma \log \sum_a \exp(Q_\theta(s, a)/\sigma)$, from which the policy $\pi_\theta(a|s)$ follows from (28). Next, a forward pass computes expected state-action visitation $\mu_\theta = \mathbb{E}_{\pi_\theta}[\sum_t \beta^t \phi(s_t, a_t)]$ via dynamic programming or Monte Carlo simulation. Finally, the gradient update $\theta \leftarrow \theta + \alpha(\mu_E - \mu_\theta)$ adjusts reward parameters; this gradient is

[18] Ziebart et al. (2008) applied this to model route choices of Yellow cab drivers in Pittsburgh, correctly predicting nearly 80% of paths in a holdout sample.

both the score of the log-likelihood under the MLE derivation and the direction that enforces the feature-matching constraint under Max-Ent. This requires solving the soft Bellman equation and computing state visitation frequencies at each iteration, making the method model-based. Classic MaxEnt IRL further requires a discrete, enumerable state space and knowledge of the transition dynamics $p(s'|s,a)$, limiting its applicability to tabular settings.

Abbeel (2016) (Guided Cost Learning, GCL) scaled MaxEnt IRL to high-dimensional continuous domains by addressing these limitations. Three key innovations enable this: (1) a neural network parameterizes the reward function $r_\theta(s,a)$, replacing hand-crafted features; (2) importance sampling corrects the distribution mismatch when policy samples come from a suboptimal generator rather than the current optimal policy; and (3) lazy policy optimization runs the inner RL loop for only a few iterations per reward update, avoiding the prohibitive cost of full convergence. GCL approximates the MaxEnt gradient $\mu_E - \mathbb{E}_{\pi_\theta}[\phi]$ using these sampled, reweighted trajectories, enabling end-to-end training in continuous control tasks. Haarnoja et al. (2017) completed the connection to DDC by deriving the soft Bellman equation from an entropy-augmented reward criterion, yielding the smoothed value function formulation discussed in Section 2.

### 4.2 Adversarial Methods

Max-Ent IRL also admits a game-theoretic interpretation via Lagrangian duality. The dual function is $g(\theta) = (\max_\pi \mathbb{E}_\pi[\theta^\top \Phi(\tau)] + H_c(\pi)) - \theta^\top \mu_E$, and the dual problem is $\min_\theta g(\theta)$. This is a saddle-point problem: the outer minimization finds reward parameters $\theta$, while the inner maximization finds the entropy-regularized optimal policy for that reward. This dual view motivates adversarial methods that implement the game explicitly. The bridge to these methods is the *discounted occupancy measure*, $\rho_\pi(s,a) = \mathbb{E}_\pi[\sum_{t=0}^\infty \beta^t \mathbb{I}(s_t = s, a_t = a)]$, which reformulates expected cumulative reward as a static inner product $\langle \rho_\pi, r\rangle$, allowing policy comparison via state-action distributions. Generative Adversarial Imitation Learning (GAIL) (Ho and Ermon, 2016) targets the full occupancy measure $\rho_\pi(s,a)$ rather than feature expectations, using a discriminator to minimize divergence to the expert's occupancy. However, the discriminator's output is entangled with dynamics and does not yield a portable reward function. The Max-Ent/MLE approaches are model-based: computing the gradient (29) requires solving for $Q_\theta$, which depends on $p(s'|s,a)$. Adversarial methods offer a scalable, model-free alternative by replacing explicit occupancy computation with a learned discriminator.

A naive adversarial IRL recovers the expert's advantage function $\mathcal{A}^*(s,a) = Q^*(s,a) - V^*(s)$, which is an "entangled" reward that is not robust to changes in dynamics since $\mathcal{A}^*$ depends on the value function $V^*$, which in turn depends on $p(s'|s,a)$. To address this, Fu et al. (2018) introduced Adversarial IRL (AIRL) to learn a *disentangled* reward. The key innovation is a structured discriminator with logits:

$$f_{\theta,\psi}(s,a,s') = g_\theta(s,a) + \beta h_\psi(s') - h_\psi(s) \tag{30}$$

This structure explicitly implements the potential-based reward shaping from Section 3.3. It is designed to decompose the learned reward into two components: a transferable, state-action reward $g_\theta(s, a)$, and a non-transferable, dynamics-dependent value function shaping term $h_\psi(s)$ that plays the role of the potential function $\Phi(s)$. Without this state-only restriction on $g_\theta$, Fu et al. (2018) show that adversarial methods recover rewards that effectively "bake in" the training dynamics $p(s'|s, a)$, causing policy failure when transferring to environments with shifted physical dynamics (e.g., a robot with modified legs). In principle, this forces $g_\theta$ to capture the intrinsic reward, enabling policy transfer to new environments with different dynamics.[19]

The AIRL algorithm proceeds iteratively. At each step, a policy $\pi$ generates trajectories by interacting with the environment. A discriminator with structure (30) is trained via binary logistic regression to distinguish expert transitions $\mathcal{D}_E$ from policy-generated ones. The trained discriminator is used to recover an updated reward signal, $r'(s, a, s') \leftarrow \log D(s, a, s') - \log(1 - D(s, a, s'))$. The policy $\pi$ is then updated using a standard RL algorithm (e.g., TRPO) to maximize this reward. This cycle of discriminating and generating improves both the reward estimate and the policy simultaneously.

The adversarial framework connects to structural econometrics: Kaji et al. (2023) show that a discriminator distinguishing real from simulated data automatically selects optimal moments, reducing to optimally-weighted SMM with a logistic discriminator. Lee et al. (2026) achieve identification by fixing the utility of an anchor action to zero within the discriminator's objective, eliminating reward shaping ambiguity and recovering interpretable preference segments from consumption trajectories.

Inverse soft-Q learning (IQ-Learn) (Garg et al., 2021) avoids the unstable adversarial game entirely. It starts from the same occupancy-matching objective but collapses the nested $\max_r \min_\pi$ problem into a single maximization over $Q$ by substituting the analytical solution $\pi_Q \propto \exp(Q)$ directly into the objective, enabling non-adversarial training. However, Kang et al. (2025) show that occupancy matching methods (including GAIL and IQ-Learn) are not guaranteed to consistently estimate the true $Q$ function: they minimize average Bellman error, not pointwise Bellman error. This implies that rewards cannot be reliably inferred from $Q$ using the Bellman equation after deriving $Q$ via occupancy matching.

### 4.3 Model-Free IRL

While principled, the methods above are model-based: they require knowledge of $p(s'|s, a)$ to compute $Q$ functions via the Bellman equation. As Adams et al. (2022) notes, "in many scenarios, the transition function may be unknown and difficult or impossible to estimate."[20] Model-free

[19] The foundational proof in Fu et al. (2018) that this decomposition successfully recovers the true reward (i.e., $g_\theta \to r$ and $h_\psi \to V$) relies on an assumption of deterministic dynamics, a key point of departure from the stochastic environments central to DDC modeling. Lee et al. (2026) demonstrate identification in stochastic settings is possible by adding further structural constraints, such as an anchor action with known utility.

[20] In RL, "model-free" means using raw transitions $(s, a, s')$ without explicitly estimating $p(s'|s, a)$, whereas "model-based" means constructing or estimating $p$ for planning (Sutton and Barto, 2020). In SE/DDC, methods using

IRL avoids this by using observed transitions $(s, a, s')$ directly, reminiscent of experience replay in deep Q-learning (Mnih et al., 2015). For example, Babes-Vroman et al. (2011) made the MLE interpretation explicit by directly maximizing trajectory likelihood, and Jain et al. (2019) make the MLE framework model-free by retaining the Max-Ent log-likelihood objective but replacing the model-based Bellman update with a differentiable variant of Q-learning, estimating gradients using only sampled transitions. We discuss two other recent contributions: AE, which can be regarded as an IRL version of the two-step CCP estimator, and GLADIUS, which is analogous to the penalized likelihood SEES estimator of Luo and Sang (2024). Both use function approximation to handle large state spaces.

Adusumilli and Eckhardt (2025) (AE) proposed a model-free estimator that combines a core method of RL, the method of *temporal differences* (TD), Sutton and Barto (2020), and the two step Hotz and Miller (1993) CCP estimator in section 3.1 that avoids the need to estimate or make parametric assumptions about $p(s'|s, a)$. Similar to the CCP estimator, they assume the reward is a linear combination of known "features", i.e. $r_\theta(s, a) = \sum_{k=1}^{K} \theta_k r_k(s, a)$ and estimate the structural parameters $\theta$ by maximizing the partial likelihood (24). However rather than numerically computing functions $R$ and $Q_\epsilon$ that enter as covariates in the CCP $\pi$ in equation (23), they use observed state transitions to estimate $R$ and $Q_\epsilon$ under the assumption they can be approximated as linear combinations of a set of known basis functions that depend on $(s, a)$ and a vector of parameters $\phi$. Let $R_\phi(s, a)$ be the approximation of $R$, and assume a similar approximation is used to compute $Q_\varepsilon$ (omitted for brevity). If $p$ were known, we could choose $\hat{\phi}$ to minimize the mean-squared TD error given by

$$\hat{\phi} = \underset{\phi}{argmin}\, E\left\{[\vec{r}(s, a) + \beta R_\phi(s', a') - R_\phi(s, a)]^2\right\}, \tag{31}$$

where $E$ denotes the expectation operator over $(s, a, s', a')$, which requires $p$. Using results from Tsitsiklis and Roy (1997) on the convergence of TD learning with linear function approximation, AE show that $\hat{\phi}$ can be computed using the *sample expectation operator* $E_N$ in (31) using the formula

$$\hat{\phi} = \left[E_N\{\nu(s, a)[\nu(s, a) - \beta\nu(s', a')]^T\}\right]^{-1} E_N\{\nu(s, a)\vec{r}(s, a)\}. \tag{32}$$

Note that relying on the sample expectation $E_N$ implicitly assumes rational expectations, as it constrains the agent's subjective beliefs to match the empirical data distribution. Using $\hat{\phi}$ the implied estimate $\hat{R}(s, a) = \phi(s, a)\hat{\phi}$ that can be used as the covariate in the CCP estimator of $\theta$ in (23), and the other "correction term" $\hat{Q}_\varepsilon(s, a)$ can be estimated in similar fashion. The asymptotic distribution of this two stage TD/CCP estimator of $\theta$ is affected by "estimation noise" in the $\hat{R}$ and $\hat{Q}_\varepsilon$ functions that can introduce bias and noise. To deal with this, AE propose a third estimation stage that uses a modified score (i.e. gradient) of the partial log-likelihood (24) to

historical data are often called "model-based" since $p$ can be estimated from observed transitions. We follow RL terminology throughout.

create a robust (Neyman orthogonal) version of the score function (Chernozhukov et al., 2022), obtaining an improved estimator $\hat{\theta}$ robust to first stage estimation noise in $\hat{R}$ and $\hat{Q}_\varepsilon$ by setting the modified score to zero.[21]

The GLADIUS estimator (Kang et al., 2025) (Gradient-based Learning with Ascent-Descent for Inverse Utility learning from Samples) is model-free, does not require first-stage CCP estimation, and uses flexible function approximation. It parameterizes $Q_{\phi_1}(s,a)$ and $EV_{\phi_2}(s,a) = \mathbb{E}[V(s')|s,a]$ using neural networks, recovering the reward as $r(s,a) = Q_{\hat{\phi}_1}(s,a) - \beta EV_{\hat{\phi}_2}(s,a)$ via the Bellman equation (13). GLADIUS maximizes a penalized likelihood where the penalty targets the Bellman error $\rho_{BE}(Q) = E\{[Q(s,a) - \Lambda_\sigma(Q)(s,a)]^2\}$. It builds on an insight of Antos et al. (2008) that the $Q$ that minimizes mean squared Bellman error does not necessarily minimize mean squared TD error given by

$$\rho_{TD}(Q_{\phi_1}) = E\left\{\left[r(s,a) + \beta\sigma \log\left(\sum_{a' \in A(s')} \exp\{Q_{\phi_1}(s',a')\}/\sigma\right) - Q_{\phi_1}(s,a)\right]^2\right\}, \tag{33}$$

where we use equation (14) to write $V(s')$ as a log-sum of $Q_{\phi_1}$. The quantities are related by:

$$\rho_{TD}(Q) = \rho_{BE}(Q) + \beta^2 E\{[V(s') - EV(s,a)]^2\} \tag{34}$$

The second term is the conditional variance of next-period values. GLADIUS addresses this via a max-min formulation: the outer maximization finds $\phi_1$ to maximize the penalized likelihood, while the inner minimization finds $\phi_2$ to estimate the conditional variance term. This targets the correct Bellman error without requiring $p(s'|s,a)$. Like the CCP estimator, GLADIUS imposes a normalizing action assumption for identification, using the Hotz-Miller inversion (27) to recover remaining $Q$ values. Under realizability assumptions, GLADIUS achieves global convergence with error $O(1/T) + O(1/N)$, the first such guarantee for methods minimizing mean squared Bellman residuals.

### 4.4 Theoretical Frontiers

A frontier of research integrates behavioral economics constraints directly into IRL. Hoiles et al. (2020), for example, model agents under Rational Inattention, where acquiring information about the state $s$ incurs an information-theoretic cost. The agent first chooses an attention policy $\alpha(s'|s)$ (mapping true states $s$ to perceived signals $s'$) to maximize expected utility net of information costs. The observed choice probability $\pi(a|s)$ emerges as a mixture over these signals. The IRL problem is then to test whether observed choices are consistent with a reward function $r_\theta(s,a)$ solving this

[21] See Khwaja and Srivastava (2025) for an alternative approach to AE's TD methods that they call RLTD-CCS that combines the TD algorithm from RL with the Conditional Choice Simulator (CCS) estimator of Hotz et al. (1994) that relies on forward simulations of paths to produce a Monte Carlo estimate the $Q_\theta(s,a)$ and hence is a model-based approach to estimation.

attention-constrained optimization. This revealed preference approach allows simultaneous recovery of the agent's reward parameters $\theta$ and their implicit information acquisition cost, providing a framework to test for and model bounded rationality directly from choice data.

Another theoretical frontier establishes fundamental limits for IRL. Metelli et al. (2023) derive the first minimax sample complexity lower bound for recovering the feasible reward set (all rewards consistent with expert behavior), proving that any algorithm requires at least $\Omega(H^3SA/\epsilon^2)$ samples. This provides a baseline for the inherent difficulty of the IRL problem.

### 4.5 Synthesis and Applications

We have surveyed IRL methods most relevant to DDC. Model-free methods avoid specifying $p(s'|s,a)$ but implicitly assume rational expectations. For counterfactual analysis involving changed dynamics, model-based methods remain necessary, since inaccurate models from finite data with limited coverage can compound errors in estimated rewards (Zeng et al., 2024). Furthermore, Schlaginhaufen and Kamgarpour (2024) demonstrate that with finite demonstration data, the theoretical rank conditions for transfer (Rolland et al., 2022) are insufficient. They propose using principal angles between the transition laws of observed experts to quantify the environment diversity required to bound the error of the recovered reward in counterfactual settings.

Ziebart et al. (2010)'s maximum causal entropy IRL, for instance, is nearly isomorphic to DDC models under the AS/CI/EV assumptions discussed in section 3. As Gleave and Toyer (2022) notes: "algorithms based on MCE IRL have scaled to high-dimensional environments. Maximum Entropy Deep IRL Wulfmeier et al. (2016) was one of the first extensions, and is able to learn rewards in gridworlds from pixel observations. More recently, Guided Cost Learning Abbeel (2016) and AIRL have scaled to continuous control tasks."

Large-scale applications include Barnes et al. (2024) using IRL to infer route preferences from Google Maps data and Zhao and Liang (2023) applying AIRL to taxi trajectories.[22] Liang et al. (2025) address the opacity of Deep IRL by introducing a post-hoc interpretation framework based on knowledge distillation. They train a surrogate Multinomial Logit (MNL) model using the "soft labels" (probability distributions) generated by the Deep AIRL policy, rather than ground-truth data. This effectively projects the high-dimensional, non-linear behavior learned by the neural network onto a linear-in-parameters structural model, allowing for the recovery of interpretable preference parameters typically associated with DDC. Lee et al. (2026) extend AIRL to unstructured text data.

Autonomous vehicles illustrate why IRL matters. Lu et al. (2023) combined imitation and reinforcement learning (BC-SAC), achieving 38% fewer safety failures than imitation alone. Meanwhile, Remonda et al. (2021) found RL agents with engineered lap-time rewards matched human speed but used fundamentally different control patterns, highlighting that engineered rewards fail

[22] The Shanghai application succeeded because road networks provide deterministic transitions and the goal was matching observed trajectories rather than counterfactual analysis.

to capture implicit human trade-offs.

### 4.6 Alternative Approaches

Closely related to IRL is the Inverse Optimal Control (IOC) tradition in control theory. Originating with Kalman (1964) for linear-quadratic systems, IOC was later adapted to complex, nonlinear biological behaviors by Mombaur et al. (2010) and Berret et al. (2011). Their approach employed numerical bilevel optimization that, analogous to the nested fixed point algorithm in DDC, requires solving the full forward control problem within the inner loop of parameter estimation. The computational intractability of this inner loop for high-dimensional continuous domains led to the local probabilistic approximations of Levine and Koltun (2012), which in turn enabled the deep, sampling-based Guided Cost Learning of Abbeel (2016), which merged the IOC tradition with IRL.

A parallel stream in cognitive science connects "mentalizing" to inverse reinforcement learning (Ng and Russell, 2000). Baker et al. (2009) modeled action understanding as inverse planning, a framework explicitly connected to the IRL literature by Jara-Ettinger (2019): observers infer unobservable mental states by inverting a forward RL model. These models extend standard IRL by recovering subjective epistemic and motivational states. The Bayesian Theory of Mind (BToM) framework treats agents as POMDP planners acting on potentially false beliefs (Baker et al., 2017), while the "naive utility calculus" (Jara-Ettinger et al., 2016) grounds social reasoning in agent-specific cost-reward trade-offs. Although recent approaches like ToMnet (Rabinowitz et al., 2018) scale via meta-learning without recovering explicit reward functions, this literature provides ways to model beliefs as separate from states and offers behavioral microfoundations for IRL.

## 5 Discussion

Our survey reveals the broad similarity between the DDC literature which arose in the 1980s and IRL which over a decade later. Their common goal is to infer a reward function $r$ presumed to govern the observed behavior of a sample of human decision makers ("demonstrations" in IRL). Both assume that humans are rational/optimizing "experts" and both use the same mathematical framework to model optimal dynamic decision making (e.g. MDPs based on "soft Q" and "softmax" formulas of section 3). However the literatures differ on the derivation and interpretation of these equations. DDC assumes that agents use pure strategies and explains apparent randomness in their behavior as due the presence of state variables they observe but outside observers don't. IRL explains randomness by assuming agents use mixed strategies which can be shown to be optimal when they care about the entropy of their strategy in addition to the rewards it generates.

The literatures also differ on the algorithms/estimation methods to infer rewards. The predominant framework in DDC is maximum likelihood, with careful attention to the identification problem and statistical properties of estimators (e.g. consistency and asymptotic distribution) for accurate inference/understanding of human behavior. This includes hypothesis tests of whether human be-

havior really is rational, and if so, to use these models for counterfactual prediction and policy evaluation (i.e. "crash testing" proposed new policies using computer simulations rather than real world implementation and trial and error). IRL is less interested in understanding human behavior and more interested in using humans as experts to train algorithms to automate complex tasks without explicit programming — such as driving a car. This likely explains the myriad of different IRL algorithms that minimize different mathematical loss functions to measure the goodness of fit between observed behavior and the predictions of the model. Nevertheless, maximum likelihood (and Bayesian inference) is predominant and closely related to two other commonly used methods in IRL: 1) maximum entropy and 2) adversarial inverse reinforcement learning. Despite differences, DDC and IRL infer rewards using the same overall approach, as a *bilevel optimization problem* with an "outer" maximization problem that searches over reward functions to maximize a measure of "model fit" (e.g. likelihood function) between observed behavior and behavior implied by an optimal policy $\pi_r$, and an "inner" minimization (or fixed point problem) that solves the MDP for the optimal policy $\pi_r$ for each trial $r$.

We discussed two key challenges facing both literatures: 1) the identification problem, and 2) the curse of dimensionality. The latter is problematic for DDC and IRL due to the need to repeatedly solve an MDP in the inside loop of the bilevel optimization problem. We discussed the use of neural networks and randomization as two ways to circumvent the curse of dimensionality. While there is no formal proof that they actually "break" this curse, the astounding successes of deep Q-learning in RL applications such as Alpha-Zero's ability to beat any human player in chess or Go suggest that these methods combined with sufficient computer power will enable us to use RL, DDC and IRL to tackle an increasing number of large scale, realistic problems. At the same time, these RL success stories call into question the assumption of human rationality: if Alpha-Zero beats all humans, we evidently cannot be perfectly rational optimizers. Instead humans may be better modeled as *boundedly rational* (Simon, 1957), but there is relatively little work on how to adapt DDC and IRL to infer rewards of boundedly rational agents and understand whether suboptimal "satisficing" behavior is due to inability to optimize or irrational beliefs.[23]

To overcome the identification problem, we have to be willing to impose strong *a priori* assumptions about rewards and beliefs, otherwise it is not possible to uniquely invert (identity) rewards and beliefs consistent with observed behavior and make valid counterfactual predictions. But this means that counterfactual predictions are dependent on hard-to-verify assumptions. This is not just a problem for the credibility of counterfactual forecasts of economic policies from DDC models: it also limits the capability of IRL to do *transfer learning,* i.e. to use rewards inferred from data gathered in one environment re-train an agent using RL to behave intelligently in a different environment with different dynamics encoded by the transition probability $p$. For example, an important use of IRL is to infer the reward function $r$ for human drivers to train autonomous

[23] Anderson et al. (2025) argue that suboptimal play of professional tennis players is due to distorted subjective beliefs, and Mu et al. (2025) show that suboptimal behavior of human subjects in lab experiments is due to subjective beliefs that differ from rational beliefs implied by Bayes Rule.

vehicles (AV). If we identify rewards using data in an urban environment where traffic patterns are captured by a transition probability $p(s'|s,a)$ and we make incorrect assumptions, IRL can mis-identify the reward function. Then even though an IRL-trained AV may perform well (e.g. human-like) in the urban environment it was trained in, the AV may perform poorly in a rural environment where traffic evolves according to a different transition probability $\lambda(s'|s,a)$. Thus, mis-identified structural models may be subject to the same problems for counterfactual prediction that we noted for reduced-form models (e.g. *behavioral cloning*): they can fail to correctly predict how behavior changes in a new environment. Despite these challenges, we know that when correct (or at least approximately correct) identifying assumptions are imposed, DDC and IRL model can make accurate counterfactual/out-of-sample predictions. For example, Todd and Wolpin (2006) showed that a structural model estimated using data on a control group accurately predicts the change in behavior of a treatment group in a randomized experiment, and Zhuang et al. (2024) showed that AVs governed by reward functions trained by IRL "can guide the direction of the policy network well in driving scenarios it had not encountered during training, thus validating the generalization capabilities of the reward functions across different drivers." (p. 8100).

Our survey also touched on the pros and cons of model-free vs model-based approaches to RL and IRL. Most of the work in DDC is model-based, i.e. it requires the researcher to specify and estimate a model for the transition probability $p(s'|s,a)$ which is estimated along with the reward function. While $p$ be estimated non-parametrically in a first stage estimation before estimating $\{\beta, r\}$, this assumes agents have rational expectations and entails the burden of numerical integration with respect to $p$ to solve the associated MDP for the optimal policy $\pi_r$. Model-free IRL methods have been developed that mimic the model-free approaches developed in the RL literature such as Q-learning discussed in section 2.4, but instead of being done in real-time in *online training* using the "environment" to simulate draws $s' \sim p(s'|s,a)$, model-free IRL can be done *offline* using historical data on transitions to fit parametric approximations (e.g. deep Q networks) $Q_\phi$, bypassing the need to estimate $p$ or do explicit numerical integration. Instead, the estimation problem is transformed into a bilevel optimization where network parameters $\phi$ are repeatedly chosen to approximate $Q$ by minimizing mean square Bellman errors in the inside loop. Thus, model-free IRL has computational advantages and avoids the need to specify and explicitly estimate $p$, but it also depends on the strong assumption that individuals have rational expectations, which can be a dubious assumption as we noted above.

Model-free methods also depend on sufficient data on $(s', s, a)$ transitions. Limited/sparse data can lead to poor estimates of $Q$ and poor implied estimates of the reward function, and model-based approaches may better leverage prior knowledge to produced better results. Even with sufficient data, such as the case of AVs where "easily captured large-scale datasets of human driving to be used to train deep learning approaches to near human standard" (Mero et al., 2022), for these safety-critical applications such as developing AVs, substantial additional online training is required over a wide range of environments before there is sufficient evidence that AVs can be trusted to

operate safely online in real time. (cite rand study on simulations necessary) But offline training via computer simulations requires a sufficiently accurate model of a wide range of driving environments.

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